

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Heavy metals: lead, mercury - Set 1

Mark scheme

AQA spec 3.4.3.2.11; Pollution; textbook pp. 275-279

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• $(15.0 - 2.2) / 15.0 \times 100$ • = 85.3% (accept 85%)	2
01.2	• Added as an anti-knock agent to smooth combustion and reduce engine wear • Lead particulates were released in exhaust gases into the air of cities and inhaled / deposited on soil and food • Lead is a neurotoxin that bioaccumulates; chronic exposure damages the nervous system / brain development in children	3
01.3	<i>Any 2 from:</i> • Replacement of lead water pipes / lead solder with copper / tin-based solder; phosphoric acid dosing of water supplies forming insoluble lead phosphate • Phasing out of lead-based paint / safer paint removal • Lead-free electrical solder • Restrictions on lead fishing weights / shotgun pellets • Industrial emission controls	2
01.4	• Reacts with the lead in old pipes to form an insoluble layer of lead phosphate on the inside of the pipe • This prevents lead dissolving into the water	2
01.5	• Swans / ducks swallow them (mistaking them for snails / grit) and the lead is absorbed, causing lead poisoning	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-276; AQA spec 3.4.3.2.11.	

Question	Marking guidance	Marks
02.1	• Inorganic mercury settled into anaerobic sediments where anaerobic microbes converted it to organic (methyl) mercury • Methyl mercury is liposoluble so it is absorbed easily and stored in fat / passes through cell membranes • It bioaccumulated in fish and shellfish and biomagnified along the food chain to reach much higher concentrations than in the water • Local people ate large amounts of fish (critical group) so received toxic doses	4
02.2	• Liposoluble so it crosses the blood-brain barrier into the brain, inhibiting enzymes / nerve function (neurotoxin) • It crosses the placenta and harms the developing unborn baby • Mercury is a teratogen, preventing normal gene expression / development	3
02.3	• Elemental (liquid) mercury: not easily absorbed through skin or gut, though vapour is absorbed in the lungs • Inorganic compounds (eg mercury oxide): moderately well absorbed in the gut • Organic compounds (eg methyl mercury): easily absorbed through skin, gut and lungs; liposoluble so most toxic and bioaccumulates	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 276-277; AQA spec 3.4.3.2.11.	
03.1	<i>Any 3 from:</i> • Disposal of items containing mercury: batteries, thermometers, fluorescent lights • Chemical plants producing chlorine using mercury electrodes • Combustion of coal • Gold mining (amalgamation)	3
03.2	<i>Any 3 from:</i> • Replacement of mercury thermometers with alcohol / electronic thermometers (discontinued uses) • Removal from effluents by reverse osmosis / activated carbon filters / ion exchange filters • Disposal / storage of wastes at high pH to reduce solubility • Encapsulation of mercury-containing hazardous waste in cement	3
03.3	• Most heavy metals are more soluble at low pH / less soluble at high pH • So at high pH they remain insoluble / immobile and cannot be leached into water or absorbed by organisms	2
03.4	• The effects of two pollutants interact to produce a different (greater) effect than either alone • Cadmium and zinc; ozone and sulfur dioxide	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-278; AQA spec 3.4.3.2.11.	
04.1	• $180 / 1.2 = 150$	1
04.2	• Cadmium and zinc occur together in the same ores so mine drainage / smelting releases both	1

Question	Marking guidance	Marks
04.3	• Mine drainage water contaminated irrigation water / paddy soil so rice absorbed cadmium ($1.2 \text{ vs } 0.05 \text{ mg kg}^{-1}$) • Cadmium is liposoluble / not excreted so it bioaccumulates in the body (kidneys) with chronic exposure to small doses over years • Cadmium causes kidney failure and bone decalcification / skeletal collapse (itai-itai disease) • Rice is the staple food so residents receive a large dose; village B has no exposure and no disease (correlation supports causation)	4
04.4	<i>Any 2 from:</i> • Collect mine drainage water and neutralise / raise pH with lime so cadmium precipitates and stays insoluble • Cover / cap spoil heaps to prevent leaching and dust • Recycle nickel-cadmium batteries / hazardous waste landfill • Phytoremediation of contaminated paddy soils	2
04.5	• Manufacture may release cadmium in effluents / dust • Panels have a limited life; disposal of old panels (landfill, incineration of pigments) could release cadmium unless recycled	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
05.1	• B	1
05.2	• TBT is highly toxic to marine molluscs / crustaceans and an endocrine disruptor altering growth and reproduction of oysters / whelks even at very low concentrations • It bioaccumulated in shellfish and could pass to humans eating them • It was assumed dilution would make it harmless (illustrates the need for the precautionary principle); now prohibited and replaced by copper	3
05.3	• In the low-oxygen conditions of the spoil heap iron is in its soluble reduced form and is leached into the river • In the river it is oxidised to insoluble iron oxide (orange sediment); the oxidation uses dissolved oxygen so the river is deoxygenated, killing aerobic organisms • Control: drainage water is passed over mesh screens where the iron oxidises and is deposited before the water reaches the river; the solid iron is removed periodically	3
05.4	<i>Any 2 from:</i> • Lead-acid car batteries (recycled) • Lead flashing on roofs / windows • Radiation shielding	2
05.5	• They inhibit enzyme function, especially the enzymes of the nervous system / nerve cells	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
06.1	• Organic (methyl) mercury	1
06.2	• Anaerobic microbes converted the inorganic mercury into organic methyl mercury • Methyl mercury is liposoluble so it is absorbed, crosses into the brain and biomagnifies along the food chain to fish eaten by people	2

Question	Marking guidance	Marks
06.3	• Most heavy metals are less soluble at high pH • So they remain immobile and are not leached into water or absorbed by organisms	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	